FORM 2

THE PATENTS ACT 1970

39 OF 1970

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THE PATENT RULES 2003

COMPLETE SPECIFICATION

(SEE SECTIONS 10 & RULE 13)

1. TITLE OF THE INVENTION

AI ENABLED MOTHERHOOD SUPPORT SYSTEM: A SMART PREGNANCY HEALTH MONITORING PLATFORM

2. APPLICANTS (S)

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3. PREAMBLE TO THE DESCRIPTION

COMPLETE SPECIFICATION

THE FOLLOWING SPECIFICATION PARTICULARLY DESCRIBES THE INVENTION AND THE MANNER IN WHICH IT IS TO BE PERFORMED

Signature Not Verified
Digitally Signed.
Name: Ankita Jain
Date: 09-Feb-2026 21:28:39
Reason: Patent Efiling
Location: DELHI

PREAMBLE TO DESCRIPTION

The following specification describes in detail the **AI Enabled Motherhood Support System: A Smart Pregnancy Health Monitoring Platform**, and explains the manner in which the invention is implemented and operated.

This invention relates to the field of **artificial intelligence–based healthcare systems**, particularly focusing on **maternal and prenatal health monitoring**. The system is designed to provide intelligent, personalized, and continuous health support to pregnant women by integrating machine learning algorithms with a web-based healthcare platform.

The invention enables real-time monitoring of maternal health parameters, trimester-wise guidance for diet, exercise, and hydration, and week-by-week embryo development tracking. It further incorporates an AI-based risk prediction module capable of analyzing medical reports to detect potential pregnancy-related complications such as anemia, thyroid disorders, gestational diabetes, and high blood pressure at an early stage.

The system also includes automated notifications, reminders for daily health routines and medical checkups, and an emergency assistance feature that allows instant connectivity with healthcare professionals. Through secure data management and intelligent analysis, the invention ensures timely medical intervention, improved maternal awareness, and enhanced safety for both mother and fetus.

The complete specification herein describes the system architecture, functional modules, methodology, and operational flow of the invention, enabling effective implementation and practical application in real-world healthcare environments.

Title:

AI Enabled Motherhood Support System: A Smart Pregnancy Health Monitoring Platform

Field of the Invention

The present invention relates to the field of **digital healthcare systems** and **artificial intelligence–based medical applications**. More particularly, the invention pertains to an **AI-enabled motherhood support and pregnancy**

health monitoring platform designed to assist expecting mothers throughout their pregnancy journey.
The invention integrates **machine learning, web technologies, and health data analytics** to provide personalized trimester-wise care, maternal health monitoring, embryo development tracking, and early prediction of pregnancy-related health risks. It further falls within the domain of **maternal and prenatal healthcare management**, offering real-time guidance, lifestyle monitoring, and emergency medical support through a single unified system.
This invention is especially applicable to **remote healthcare, preventive medicine, and smart health monitoring solutions**, aimed at improving maternal and fetal health outcomes by enabling early detection of complications and timely medical intervention.

Background

Pregnancy is a critical and sensitive phase in a woman's life that requires continuous medical attention, proper lifestyle management, and timely health monitoring to ensure the safety of both the mother and the developing fetus. Despite advances in medical science, many expecting mothers still face difficulties in accessing consistent, personalized, and reliable healthcare guidance throughout their pregnancy.

Traditional pregnancy care systems are often dependent on periodic hospital visits, which may not be sufficient to detect early signs of complications such as anemia, thyroid disorders, gestational diabetes, or high blood pressure. In many cases, these conditions remain unnoticed until they reach a severe stage, increasing the risk of adverse outcomes for both mother and child. Additionally, generic advice provided through conventional methods does not account for individual health conditions, pregnancy stages, or lifestyle factors.

With the growing adoption of digital technologies, there is an increasing demand for intelligent healthcare solutions that can provide real-time monitoring, personalized guidance, and early risk detection. However, existing digital pregnancy applications largely focus on basic tracking or informational content and lack advanced artificial intelligence capabilities for medical report analysis, risk prediction, and emergency support.

Therefore, there exists a need for a comprehensive, AI-driven pregnancy healthcare system that integrates personalized care planning, continuous health monitoring, early detection of potential complications, and instant medical assistance. The present invention addresses these limitations by offering an intelligent, user-friendly platform that supports expecting mothers throughout their pregnancy journey while enabling timely medical intervention and improved maternal and fetal health outcomes

Objects of the Invention

The principal object of the present invention is to provide an **AI-enabled motherhood support system** that ensures safe, healthy, and well-monitored pregnancy care for expecting mothers.

Another object of the invention is to offer **personalized, trimester-wise guidance** for diet, exercise, and water intake based on the individual health condition and pregnancy stage of the mother.

Another object of the present invention is to enable **early detection and prediction of pregnancy-related health risks** such as anemia, thyroid disorders, gestational diabetes, and high blood pressure through the use of artificial intelligence and machine learning algorithms.

Another object of the invention is to provide a **real-time pregnancy monitoring platform** that tracks maternal health parameters and embryo development on a week-by-week basis.

Another object of the invention is to deliver **automated notifications and reminders** for daily health routines, medical checkups, hydration, and nutrition to ensure consistent care throughout pregnancy.

Another object of the present invention is to incorporate an **emergency assistance feature** that allows instant connectivity between the pregnant woman and qualified healthcare professionals during critical situations.

Another object of the invention is to create a **secure and user-friendly digital healthcare platform** that enables doctors and healthcare providers to remotely monitor patient data and intervene in a timely manner.

Another object of the invention is to reduce pregnancy-related complications, stress, and uncertainty by providing continuous support, guidance, and medical awareness through a single integrated system.

5 **Flowchart:**

AI Enabled Motherhood Support System
Step 1: Sign Up & Enter Details
• Create account
• Enter pregnancy month, due date, and basic health info.
Step 2: Get Personalized Dashboard
• See month-wise plan for safe exercise, healthy diet tips, and water intake tracking
Step 3: Upload Medical Reports
• AI analyzes your medical reports to spot risks like thyroid, high BP, anemia, gestational diabetes
Step 4: Monitor Baby's Growth
• View week-by-week embryo development with clear visuals and descriptions.
Step 5: Follow Daily Health Routine
• Get reminders for meals, hydration, safe exercises, and doctor appointments
Step 6: Receive Alerts & Reminders
• Get notified for important tasks to maintain health and stay on track.
Step 7: Get Emergency Help
• Press one-tap emergency button to instantly connect with a qualified doctor.
Stay Healthy & Safe
Personalized care ensures a safe, healthy, and stress-free pregnancy journey.

Brief Description of the Flowchart:

The accompanying drawing illustrates the **flowchart of the AI Enabled Motherhood Support System**, showing the overall working and operational sequence of the invention. The flowchart represents the step-by-step process starting from user registration and entry of pregnancy-related details, followed by personalized dashboard generation, medical report upload, and AI-based health risk analysis.

The drawing further depicts embryo growth monitoring, daily health routine management through reminders and alerts, and an emergency assistance mechanism that enables instant communication with healthcare professionals. The flowchart clearly demonstrates how the system integrates user inputs, artificial intelligence analysis, and healthcare services to provide continuous and reliable pregnancy support.

Detailed Description:

The AI Enabled Motherhood Support System has been designed as a complete, intelligent, and easy to use platform to help expecting mothers manage their health and pregnancy journey with confidence. It combines modern web development with machine learning to provide real-time guidance, health tracking, and early risk prediction.

Frontend (User Interface):
Built using React.js, HTML, and CSS, the interface is clean, responsive, and user-friendly. Every feature from viewing the pregnancy dashboard to tracking water intake or checking embryo growth is designed so mothers can access it comfortably from any device without any technical hassle.

Backend (Server Side):
Developed with Node.js and Express.js, the backend ensures smooth performance, secure handling of user data, and seamless communication between the frontend and the database. It handles all the core operations like authentication, data processing, and API calls efficiently.

AI Module:
The heart of the system is its Machine Learning model built in Python using Pandas and Scikit learn. We use the Random Forest

algorithm to analyze uploaded medical reports and predict possible pregnancy related health risks such as thyroid problems, anemia, gestational diabetes, or high blood pressure allowing these risks to be detected early before they become serious.

Database:
All user profiles, health history, water intake logs, embryo growth records, and AI predictions are stored securely in MongoDB, ensuring fast retrieval and safe data management.

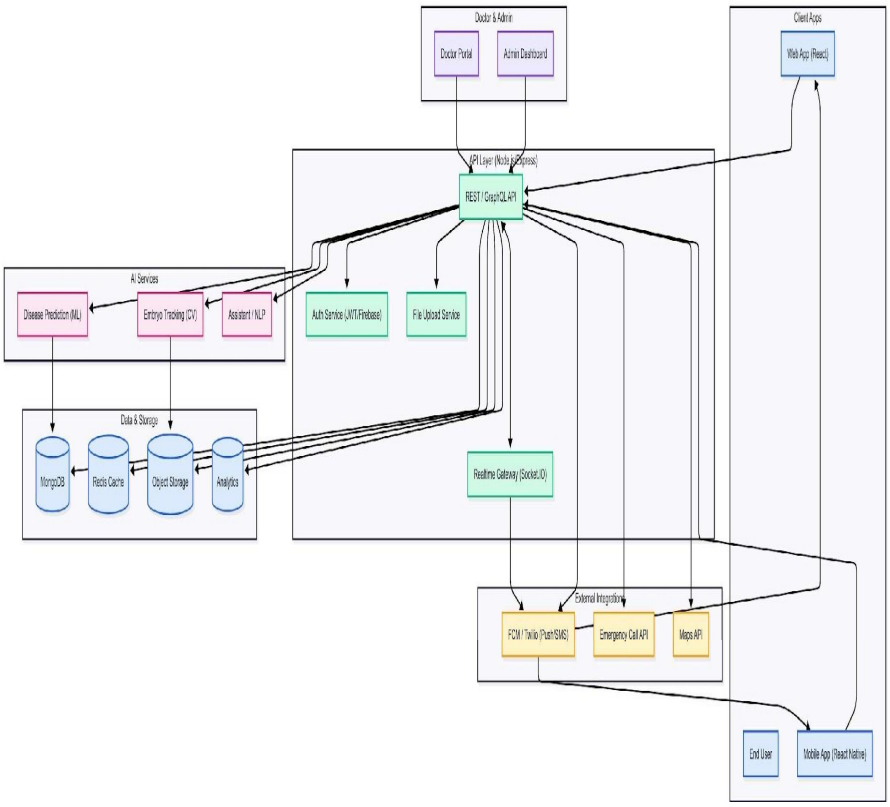
Key Modules in the System:

1. Authentication System – Secure login/registration for both mothers and doctors.
2. Pregnancy Dashboard – Displays the pregnancy month, baby's embryo stage, and important health stats.
3. Exercise & Diet Module – Gives trimester specific safe exercise routines and diet recommendations.
4. Water Tracker – Logs daily water intake and sends timely hydration reminders.
5. Embryo Development Tracker – Shows week by week visual and text updates of the baby's growth.
6. AI Disease Prediction – Predicts possible health risks based on uploaded reports.
7. Notification System – Sends reminders for meals, exercise, hydration, and checkups.
8. Emergency Call – One-tap button to connect instantly with a qualified doctor.
9. Doctor Dashboard – Lets doctors view patient history, reports, and AI alerts for timely intervention.
10. Admin Panel (Optional) – For user management, health tip broadcasts, and system analytics.

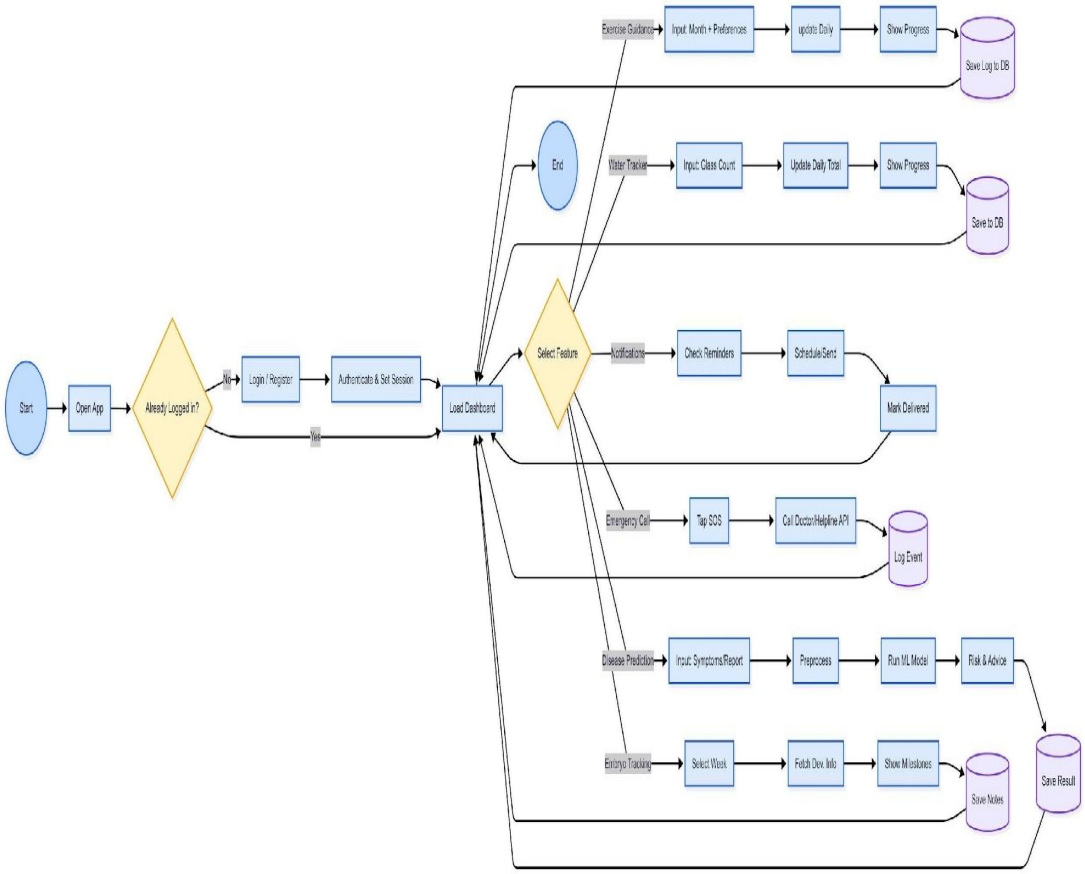
How it Works (Flow):

The process is simple, the mother enters her details and uploads any relevant medical reports → the AI model analyzes the data and updates the dashboard → personalized exercise, diet, and hydration plans are provided → reminders keep her on track → if any abnormality is detected, alerts are sent immediately and she can connect with a doctor instantly.

In short, the system works like a 24/7 personal pregnancy health assistant, making sure mothers feel safe, informed, and supported at every step.



TECHNICAL ARCHITECTURE



FLOWCHART OF SYSTEM WORKFLOW

We Claim that:

1. **An AI-enabled motherhood support system**, comprising a secure web-based and/or mobile-based digital platform, configured to provide continuous pregnancy health monitoring, personalized trimester-wise care planning, and intelligent decision support to expecting mothers throughout the entire pregnancy period.
2. The system as claimed in claim 1, wherein the platform is adapted to receive and store user-specific information including pregnancy month, expected delivery date, age, medical history, and basic health parameters, and to process the said information to generate individualized pregnancy care recommendations.
3. The system as claimed in claim 1, wherein an **artificial intelligence and machine learning module** is configured to analyze uploaded medical reports and health data using predictive algorithms in order to identify and forecast potential pregnancy-related complications such as anemia, thyroid imbalance, gestational diabetes, hypertension, and other maternal health risks at an early stage.
4. The system as claimed in claim 1, wherein the platform includes a **fetal or embryo development tracking module** that displays week-by-week growth information of the fetus, along with explanatory visual and textual content, enabling the expecting mother to understand developmental milestones during pregnancy.
5. The system as claimed in claim 1, wherein the platform incorporates a **lifestyle management module** configured to provide trimester-specific recommendations related to nutrition, diet planning, safe physical exercises, and daily water intake, thereby supporting healthy maternal habits.
6. The system as claimed in claim 1, wherein an **automated notification and reminder system** is provided to alert the user regarding scheduled meals, hydration goals, exercise routines, medical checkups, diagnostic tests, and other essential pregnancy-related activities in real time.
7. The system as claimed in claim 1, wherein the platform includes an **emergency assistance mechanism** comprising a one-tap emergency feature that enables instant communication between the expecting mother and a qualified healthcare professional for immediate medical guidance during critical situations.
8. The system as claimed in claim 1, wherein a **doctor or healthcare provider interface** is provided to allow authorized medical professionals to remotely access patient health data, AI-generated risk predictions, medical reports, and historical records for timely diagnosis and intervention.
9. The system as claimed in claim 1, wherein all user data including personal details, medical records, activity logs, AI predictions, and communication history are securely stored, encrypted, and managed within a cloud-based or distributed database infrastructure to ensure data privacy and integrity.
10. The system as claimed in claim 1, wherein the platform operates as a **24/7 intelligent digital pregnancy health assistant**, reducing dependency on

frequent physical hospital visits, minimizing pregnancy-related risks, improving maternal awareness, and ensuring safer and healthier outcomes for both the mother and the developing fetus.

Dated: 23/01/2026

NAME AND SIGNATURE OF PATENT AGENT

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APPLICANT'S PATENT ATTORNEY

TO,

THE CONTROLLER OF PATENTS

Abstract

The AI Enabled Motherhood Support System is a smart, AI powered web platform created to guide and support expecting mothers throughout their pregnancy. It provides personalized trimester wise plans for diet, exercise, and water intake, helping mothers stay healthy at every stage. The system also tracks embryo growth week by week, predicts possible health risks using AI, and sends real time reminders for daily routines and checkups.

In case of emergencies, it offers instant medical connectivity, allowing mothers to reach a qualified doctor with just one tap. By combining technology with care, the platform ensures a safer, healthier, and more confident pregnancy journey for both mother and baby.